

CMAT

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Abstract

This document provides a template for reports in the "AI in Financial Services" course, using EB Garamond for prose and Libertinus Math for formulas. It includes a cover page, abstract, table of contents, and sample sections for math and text. Additional content demonstrates tables, code, and references.

Contents

1	Overview	3
2	What is CMAT?	4
3	Exploratory Data Analysis	5
3.1	How many people went to CMAT?	5
4	Are they better?	6
4.1	How do we quantify that?	6
4.2	Then, how do we compare it?	7
5	Califications	8
5.1	Why are there missing values?	8
5.2	Imputation of missing values	8
5.3	The califications are relatable?	8
5.4	How does the teachers calificate their students?	10
6	Which is the best way to estandarize?	14
6.1	The califcotions depends on the teacher	14
6.2	The califications depend on the teacher at time t	15
6.3	Then, by salon	15
7	Analysis by calification	16
7.1	Difference of means	16
7.2	Confidence Intervals	17
7.3	Robust tests	17
7.4	Confidence intervals	18
7.5	Interpretation	18
8	Analysis by student	20
8.1	Difference of means	20
8.2	Confidence Intervals	21
8.3	Robust tests	21
8.4	Confidence intervals	23
9	Cute plot	24
9.1	Difference of means	25
9.2	Confidence Intervals	26
9.3	Robust tests	26
9.4	Confidence intervals	27
10	Conclusions	28
10.1	Context and Variables	28
10.2	Robust Nonparametric Comparison	28
10.2.1	Medians and Median Difference	28

10.2.2	Permutation Test for the Median	28
10.2.3	Mann–Whitney U and Common-Language Effect Size	29
10.2.4	Rank-Biserial Correlation	29
10.2.5	Brunner–Munzel Test	29
10.2.6	Cliff’s Delta	29
10.2.7	Mean-Based Comparison (t-Test and Cohen’s <i>d</i>)	30
10.3	Integrated Interpretation	30
10.4	Limitations and Causal Interpretation	31
10.5	Concise Quantitative Summary	31

1 Overview

This document is a minimal example using EB Garamond for prose and libertine¹ for formulas. Links like [this one](#) are active.

2 What is CMAT?

Centro de Matematicas, (Math Center), is a place where students go to take asesorias.

3 Exploratory Data Analysis

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3.1 How many people went to CMAT?

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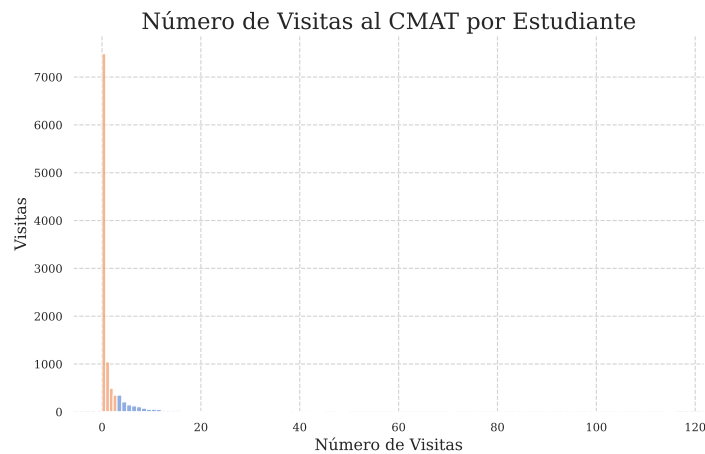


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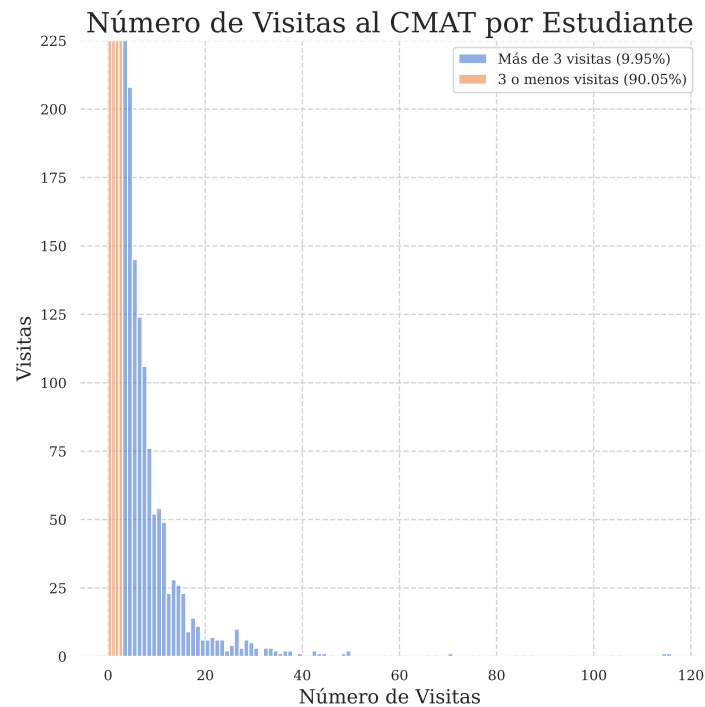


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4 Are they better?

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4.1 How do we quantify that?

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4.2 Then, how do we compare it?

Sup: With their califications.

5 Califications

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5.1 Why are there missing values?

Students have the option to do not receive calibration.

Sup: Students don't want to receive calibration when they will not pass the course.

5.2 Imputation of missing values

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5.3 The califications are relatable?

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Special cases

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Comparación de Métodos de Imputación de Calificaciones para un Salón

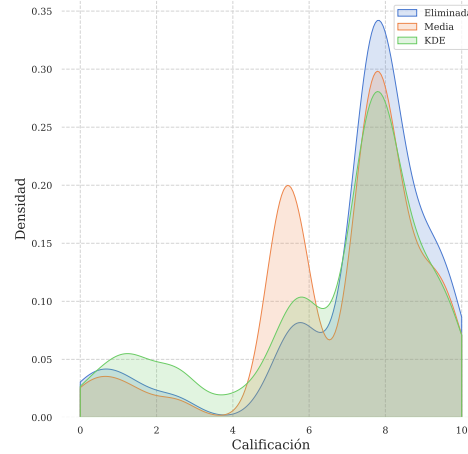


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5.4 How does the teachers calificate their students?

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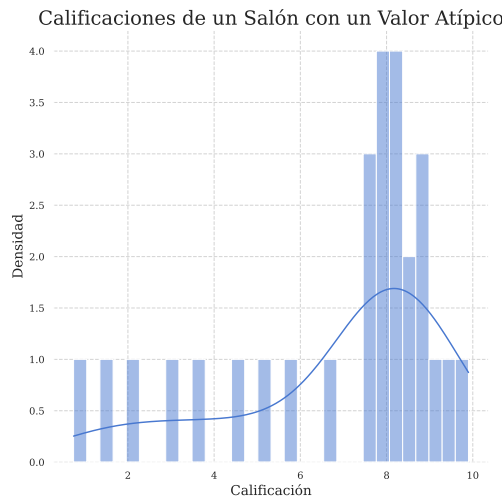


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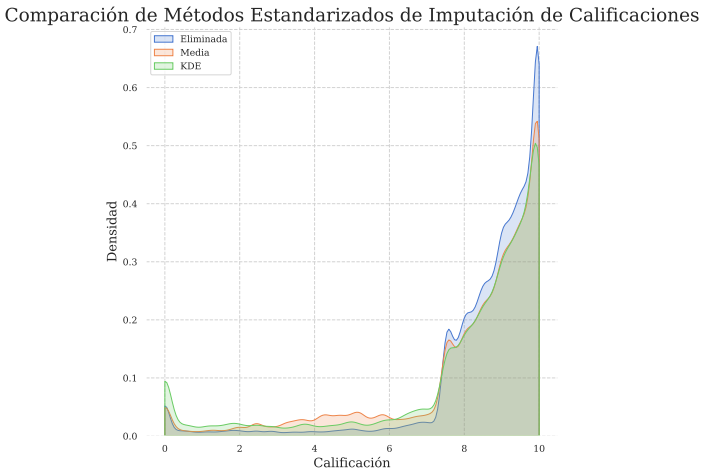


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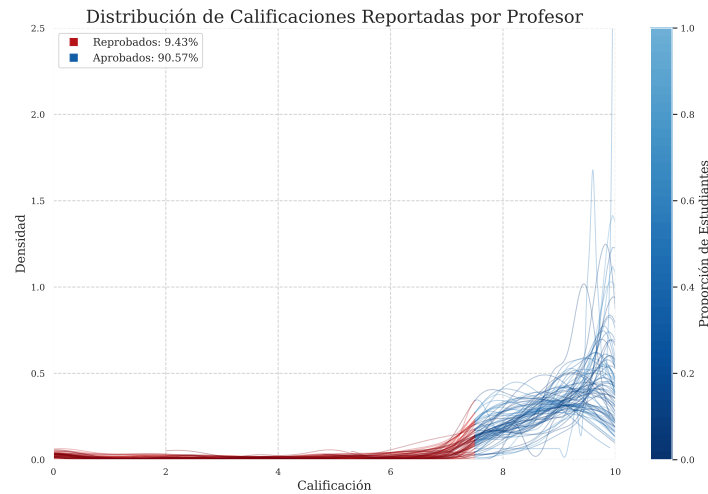


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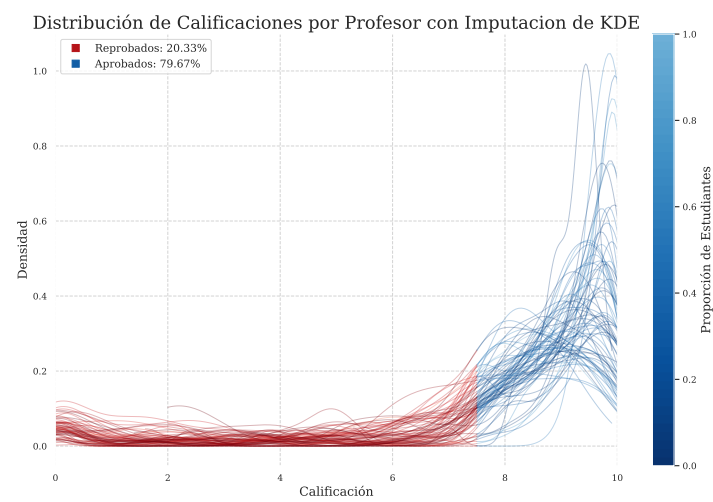


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Answer: There are 3 ways of califications

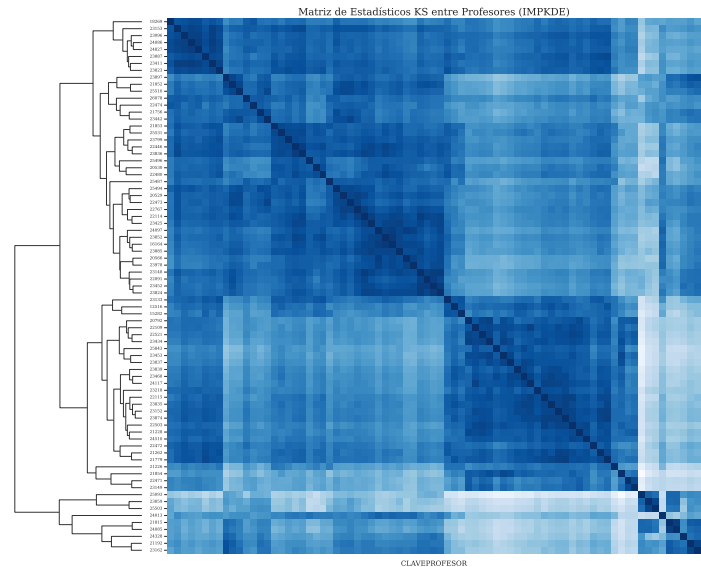


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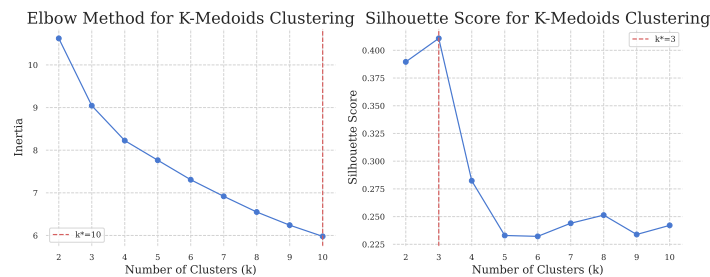


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6 Which is the best way to estandarize?

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6.1 The calificotions depends on the teacher

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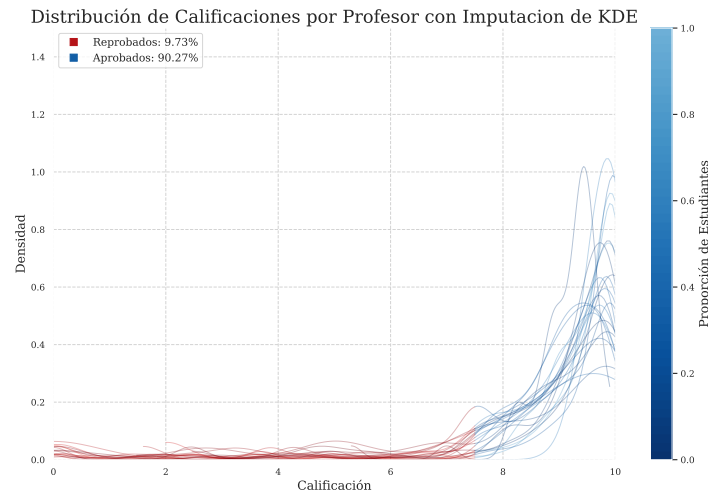


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6.2 The califications depend on the teacher at time t

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6.3 Then, by salon

We standarize by classroom.

Hooray! We got it.

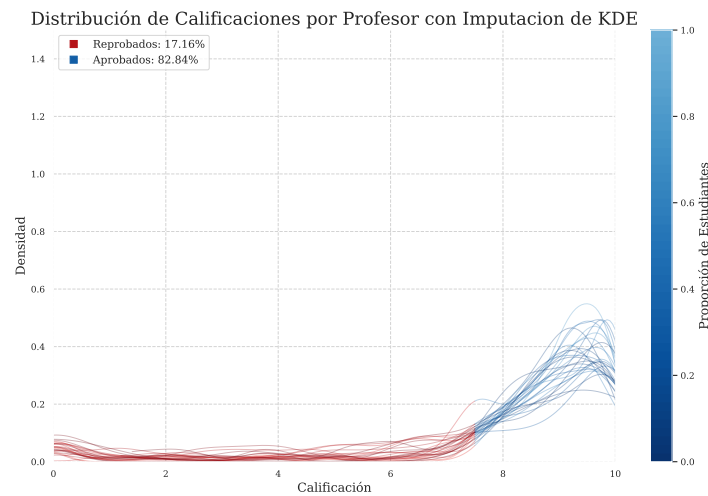


Figure 5.4.4: PDF imported as image

7 Analysis by calification

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7.1 Difference of means

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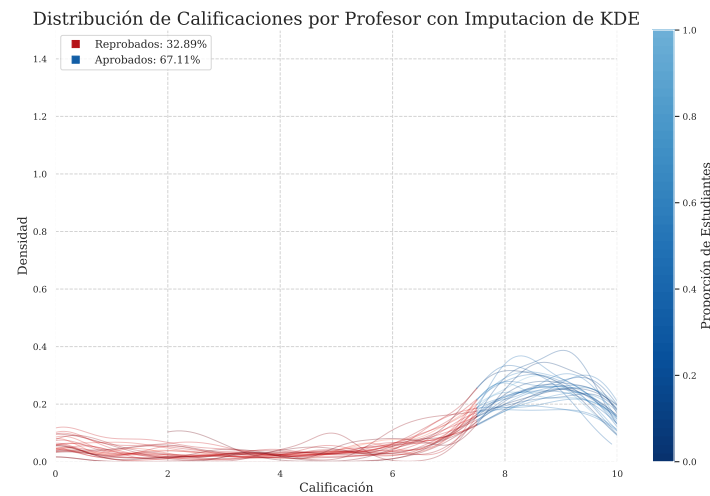


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7.2 Confidence Intervals

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7.3 Robust tests

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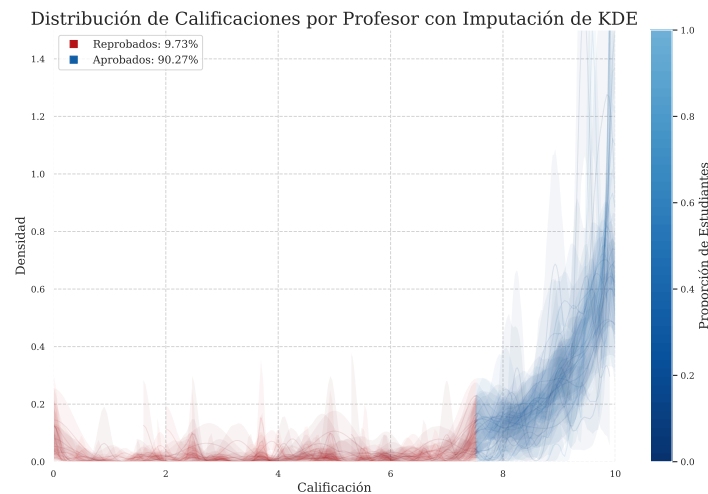


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7.4 Confidence intervals

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7.5 Interpretation

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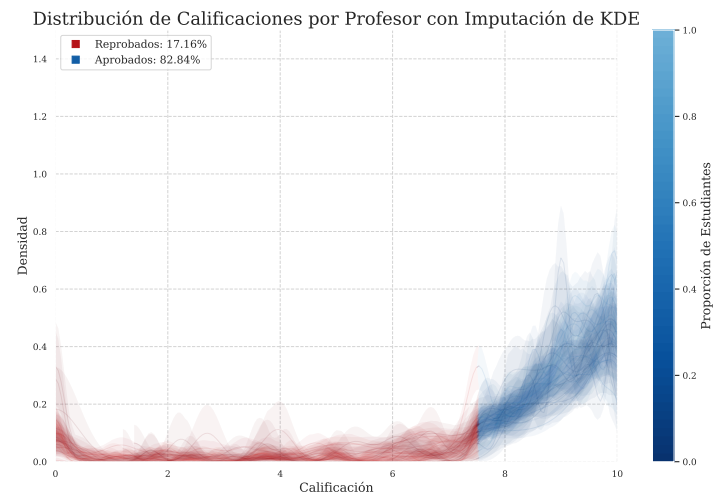


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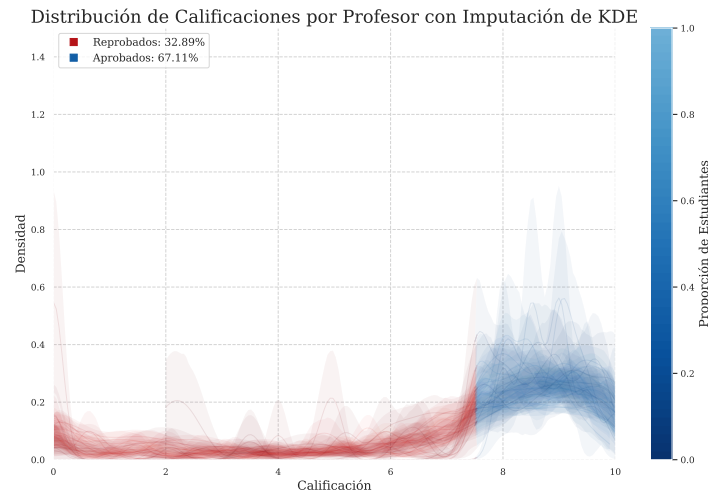


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8 Analysis by student

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8.1 Difference of means

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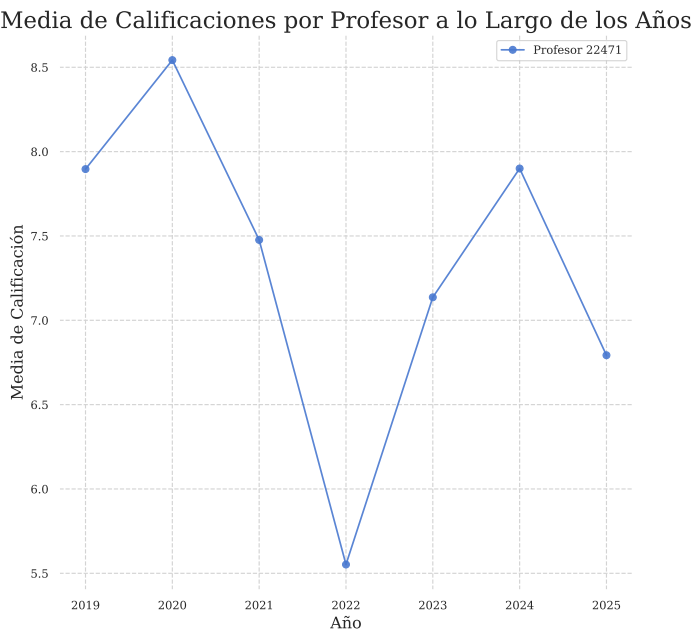


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8.2 Confidence Intervals

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8.3 Robust tests

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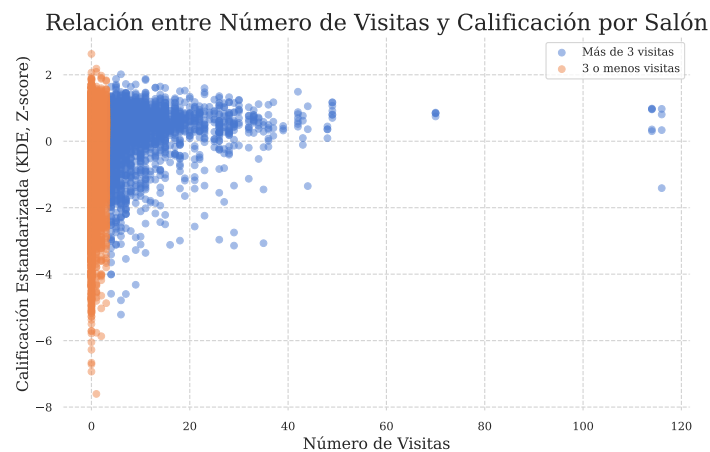


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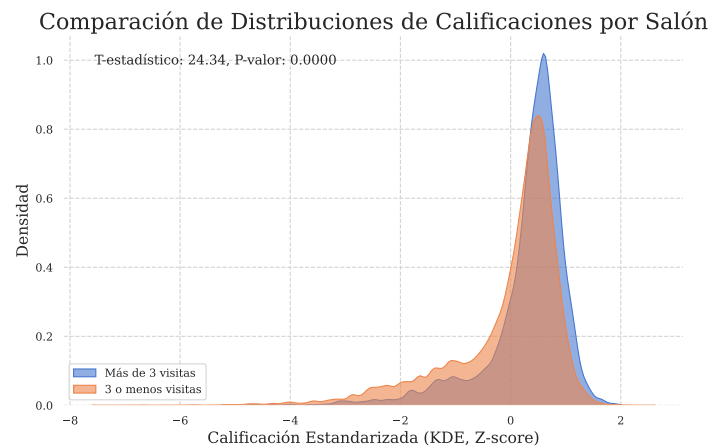


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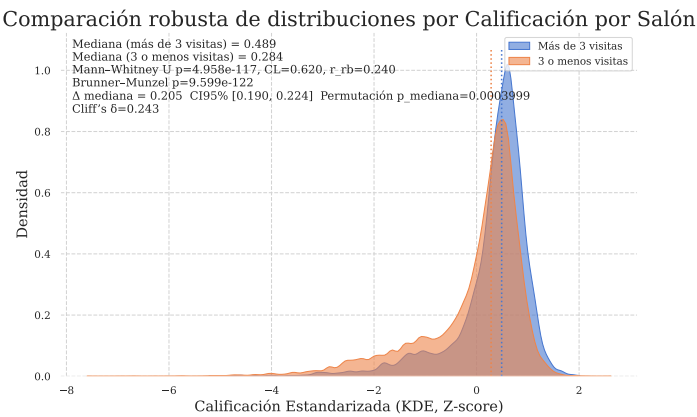


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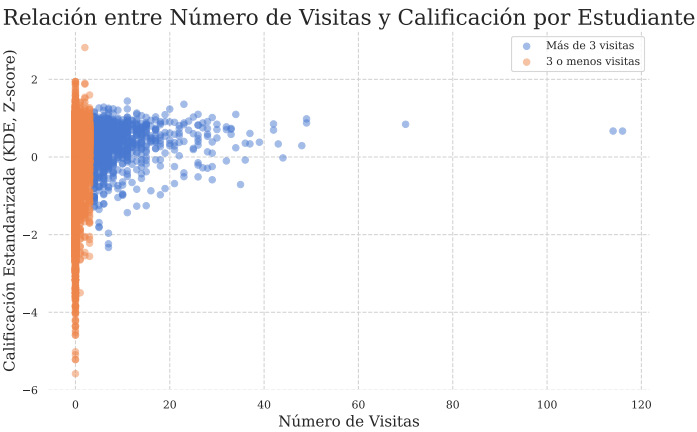


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8.4 Confidence intervals

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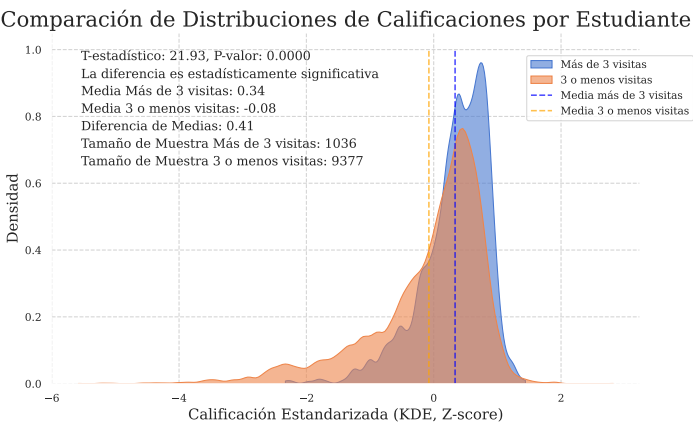


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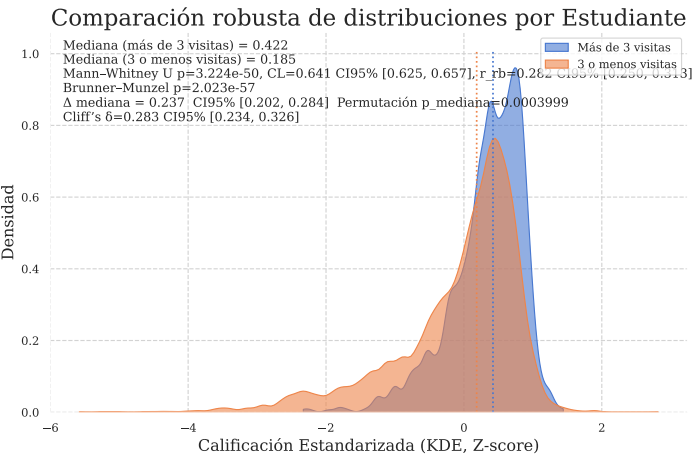


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9 Cute plot

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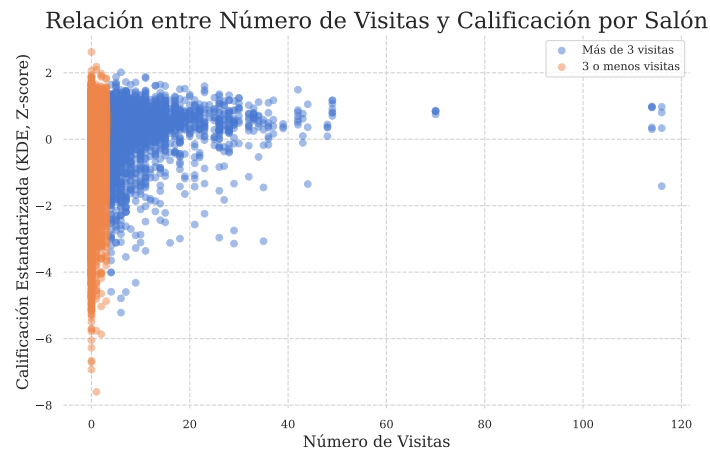


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9.1 Difference of means

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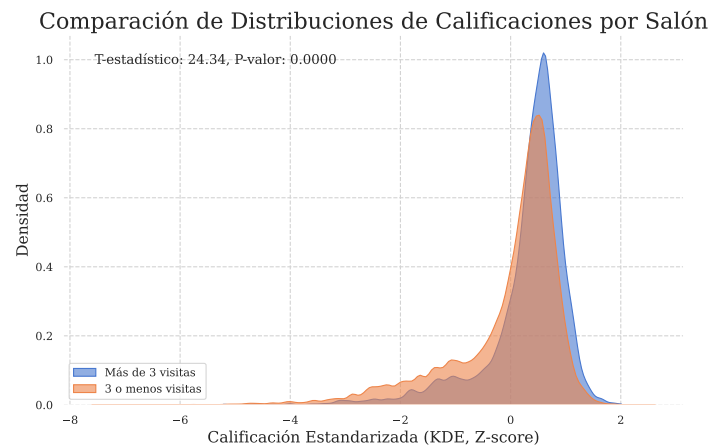


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9.2 Confidence Intervals

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9.3 Robust tests

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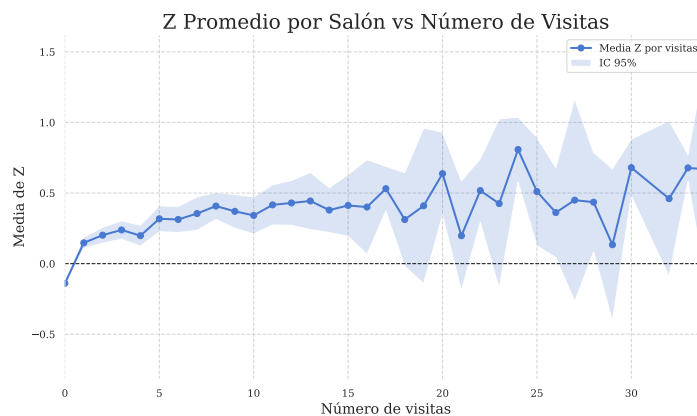


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9.4 Confidence intervals

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10 Conclusions

10.1 Context and Variables

The outcome variable is the standardized grade per student, `MEAN_IMP KDE_Z`. For each classroom, raw grades were standardized to a Z-score and missing values were imputed using kernel density estimation (KDE). All analyses are therefore carried out on a common standardized scale within classrooms.

Students were divided into two groups according to their cumulative use of the CMAT (Mathematics Learning Center):

- **Group A (frequent CMAT):** students with more than three recorded visits to CMAT ($n_A = 1036$).
- **Group B (infrequent / non-CMAT):** students with three or fewer recorded visits ($n_B = 9377$).

The goal is to quantify the association between frequent CMAT attendance and standardized grades. Because the distribution of standardized grades is non-normal and includes imputed values, the primary analyses rely on robust nonparametric methods, complemented by classical mean-based comparisons.

10.2 Robust Nonparametric Comparison

10.2.1 Medians and Median Difference

The sample medians are

$$\text{Median}_A = 0.422, \quad \text{Median}_B = 0.185.$$

Thus, a typical frequent CMAT student is about 0.42 standard deviations above the classroom mean, whereas a typical infrequent or non-CMAT student is about 0.19 standard deviations above the classroom mean.

The difference in medians is

$$\Delta_{\text{med}} = \text{Median}_A - \text{Median}_B = 0.237.$$

A nonparametric bootstrap with $B = 8000$ resamples yields a 95% confidence interval for the median difference of

$$\Delta_{\text{med}} \in [0.202, 0.284].$$

This interval lies entirely above zero, indicating that a null hypothesis of no difference in medians is not compatible with the data.

Substantively, frequent CMAT students tend to lie roughly one quarter of a standard deviation higher on standardized grades than their peers.

10.2.2 Permutation Test for the Median

A two-sample permutation test was conducted for the median difference. Under the null hypothesis that CMAT frequency has no effect and that group labels (more than three vs. three or fewer visits) are exchangeable, the observed statistic is $\Delta_{\text{med}} = 0.237$.

Based on 5000 random relabelings, the two-sided permutation p-value is

$$p_{\text{perm, median}} \approx 0.0004.$$

Hence, if the null hypothesis were true, a median difference as large as the one observed would occur in only about 0.04% of random permutations. This provides strong evidence that the two groups differ in central tendency.

10.2.3 Mann–Whitney U and Common-Language Effect Size

The Mann–Whitney U test compares the distributions using rank information. The test of equality of distributions yields

$$p_{\text{MW}} = 3.224 \times 10^{-50},$$

which is effectively zero at any conventional significance level. Thus, the null hypothesis of identical distributions is strongly rejected.

The associated common-language effect size (CL) is estimated as

$$\text{CL} = 0.641,$$

with a 95% confidence interval of

$$\text{CL} \in [0.625, 0.657].$$

This quantity can be interpreted as the probability that a randomly selected frequent CMAT student has a higher standardized grade than a randomly selected infrequent/non-CMAT student, after taking the more favorable direction. If the two groups were identical, CL would equal 0.5. The estimate of approximately 0.64 with a narrow confidence interval well above 0.5 indicates a consistent advantage for frequent CMAT attendees.

10.2.4 Rank-Biserial Correlation

The rank-biserial correlation derived from the Mann–Whitney test is

$$r_{\text{rb}} = 0.282,$$

with 95% confidence interval [0.250, 0.313]. This effect size is correlation-like and ranges from -1 to 1 . Conventional benchmarks in the behavioral sciences regard absolute values around 0.1 as small, 0.3 as moderate, and 0.5 or larger as large. Thus, $r_{\text{rb}} \approx 0.28$ corresponds to a small-to-moderate upward shift in the distribution of grades for frequent CMAT students.

10.2.5 Brunner–Munzel Test

The Brunner–Munzel test is a rank-based method that is robust to heteroscedasticity and differences in distributional shape between groups. The two-sided test yields

$$p_{\text{BM}} = 2.023 \times 10^{-57},$$

again providing overwhelming evidence against the null hypothesis of equal distributions. The agreement between Mann–Whitney and Brunner–Munzel suggests that the main findings are not driven by differences in variance or skewness alone.

10.2.6 Cliff's Delta

Cliff's delta is another nonparametric effect size summarizing stochastic dominance:

$$\delta = P(X > Y) - P(Y > X),$$

where X is a grade from the frequent CMAT group and Y is a grade from the infrequent/non-CMAT group. The estimate is

$$\hat{\delta} = 0.283,$$

with 95% confidence interval

$$\delta \in [0.234, 0.326].$$

Thus, the probability that a frequent CMAT student outperforms an infrequent/non-CMAT student exceeds the opposite probability by approximately 28 percentage points. As with r_{rb} , this is typically interpreted as a small-to-moderate effect size.

10.2.7 Mean-Based Comparison (t-Test and Cohen's d)

To complement the robust analyses, a classical comparison of means was performed. The sample means are

$$\bar{x}_A = 0.34, \quad \bar{x}_B = -0.08.$$

The difference in means is therefore

$$\bar{x}_A - \bar{x}_B = 0.41 \text{ Z-units.}$$

A two-sample t-test (assuming equal or Welch-corrected variances) produced a test statistic of

$$t \approx 21.93,$$

with approximately $n_A + n_B - 2 \approx 10,411$ degrees of freedom. The associated p-value is much smaller than 10^{-10} (on the order of 10^{-100}), again indicating that the null hypothesis of equal means is incompatible with the observed data.

From the reported t and difference in means, the pooled standard deviation can be reconstructed as $s_p \approx 0.57$. The standard errors of the group means are then

$$SE_A = \frac{s_p}{\sqrt{n_A}} \approx 0.0177, \quad SE_B = \frac{s_p}{\sqrt{n_B}} \approx 0.0059.$$

Approximate 95% confidence intervals are

$$\bar{x}_A \in [0.31, 0.37], \quad \bar{x}_B \in [-0.092, -0.068],$$

and for the mean difference

$$\bar{x}_A - \bar{x}_B \in [0.37, 0.45].$$

Thus, on average, frequent CMAT students are between 0.37 and 0.45 standard deviations above infrequent/non-CMAT students on the standardized scale.

The standardized mean difference (Cohen's d) is

$$d = \frac{\bar{x}_A - \bar{x}_B}{s_p} \approx \frac{0.41}{0.57} \approx 0.72.$$

Using conventional benchmarks (0.2 small, 0.5 medium, 0.8 large), this represents a medium-to-large effect, consistent with a substantively meaningful difference in educational outcomes.

10.3 Integrated Interpretation

Taken together, the nonparametric and parametric analyses lead to a coherent picture:

- **Direction.** All measures—median difference, common-language effect size, rank-biserial correlation, Cliff's delta, and the difference in means—indicate that students who visit CMAT more than three times tend to achieve higher standardized grades than those who visit three or fewer times.
- **Magnitude.** In terms of central tendency, the median for frequent CMAT students is about 0.24 standard deviations higher, and the mean is about 0.41 standard deviations higher. In pairwise terms, frequent CMAT students are expected to outperform infrequent/non-CMAT students roughly 64% of the time, with an excess win probability of about 28 percentage points.
- **Statistical evidence.** All p-values from the Mann–Whitney, Brunner–Munzel, permutation, and t-tests are extremely small. The findings are not marginally significant; they represent very strong statistical evidence of association.

- **Robustness.** The conclusions are supported both by distribution-free methods (ranks, medians, Cliff’s delta, bootstrap) and by classical mean-based inference, suggesting that the result is not an artifact of non-normality or heteroscedasticity.

In summary, frequent CMAT attendance is strongly and consistently associated with improved standardized performance. As W. Edwards Deming is often paraphrased: “*In God we trust; all others must bring data.*” In this dataset, the data indicate a clear and robust performance advantage for students who engage more intensively with the CMAT.

10.4 Limitations and Causal Interpretation

Despite the strength of the association, these analyses are based on observational data and do not, by themselves, establish a causal effect of CMAT attendance on grades. Several caveats apply:

- **Self-selection.** Students with higher prior ability, greater motivation, or other unobserved advantages may be more likely to use CMAT frequently. In that case, part of the observed effect reflects selection into CMAT rather than the impact of CMAT itself.
- **Unobserved confounding.** The standardization by classroom and KDE-based imputation control some aspects of the data-generating process, but not others, such as prior GPA, socioeconomic status, or access to private tutoring.
- **Imputation assumptions.** KDE-based imputation assumes that missingness mechanisms are not systematically different between groups. If missing grades are more likely in one group for reasons related to performance, the imputation may introduce bias.

For these reasons, the most defensible substantive claim is that frequent CMAT attendance is *strongly associated* with higher standardized performance. Demonstrating that CMAT causes higher grades would require additional causal design elements, such as longitudinal models, matching or weighting on baseline covariates, or experimental interventions.

10.5 Concise Quantitative Summary

The core quantitative results can be summarized as

$$\Delta_{\text{med}} \approx 0.24 \text{ Z-units (95\% CI [0.20, 0.28])}, \quad \text{CL} \approx 0.64, \quad \delta \approx 0.28, \quad d \approx 0.72.$$

Frequent CMAT users outperform infrequent or non-users with a small-to-moderate but robust advantage in standardized grades.